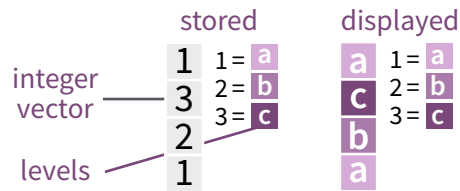


Factors with forcats : : CHEATSHEET

The **forcats** package provides tools for working with factors, which are R's data structure for categorical data.

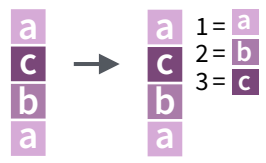
Factors

R represents categorical data with factors. A **factor** is an integer vector with a **levels** attribute that stores a set of mappings between integers and categorical values. When you view a factor, R displays not the integers, but the levels associated with them.



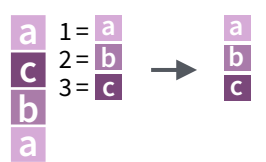
Create a factor with `factor()`

factor(x = character(), levels, labels = levels, exclude = NA, ordered = is.ordered(x), nmax = NA) Convert a vector to a factor. Also **as_factor()**.
f <- **factor**(c("a", "c", "b", "a"), levels = c("a", "b", "c"))



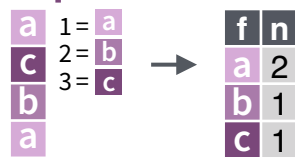
Return its levels with `levels()`

levels(x) Return/set the levels of a factor. `levels(f)`; `levels(f) <- c("x", "y", "z")`

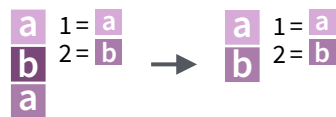


Use `unclass()` to see its structure

Inspect Factors



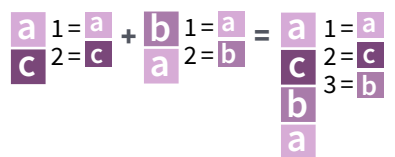
fct_count(f, sort = FALSE, prop = FALSE) Count the number of values with each level. `fct_count(f)`



fct_match(f, lvls) Check for lvls in f. `fct_match(f, "a")`

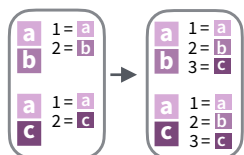
fct_unique(f) Return the unique values, removing duplicates. `fct_unique(f)`

Combine Factors



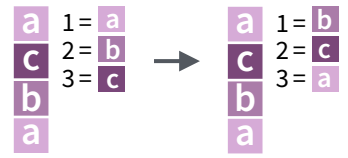
fct_c(...) Combine factors with different levels. Also **fct_cross**().

f1 <- **factor**(c("a", "c"))
f2 <- **factor**(c("b", "a"))
fct_c(f1, f2)



fct_unify(fs, levels = lvls_union(fs)) Standardize levels across a list of factors. `fct_unify(list(f2, f1))`

Change the order of levels



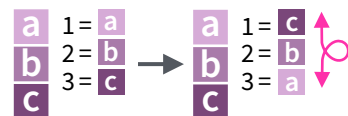
fct_relevel(f, ..., after = 0L)
Manually reorder factor levels.
`fct_relevel(f, c("b", "c", "a"))`



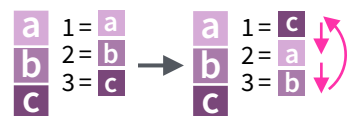
fct_infreq(f, ordered = NA) Reorder levels by the frequency in which they appear in the data (highest frequency first). Also **fct_inseq()**.
f3 <- **factor**(c("c", "c", "a"))
`fct_infreq(f3)`



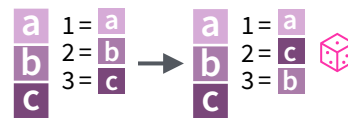
fct_inorder(f, ordered = NA)
Reorder levels by order in which they appear in the data.
`fct_inorder(f2)`



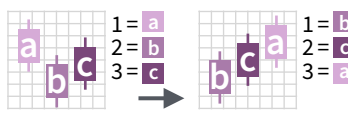
fct_rev(f) Reverse level order.
f4 <- **factor**(c("a", "b", "c"))
`fct_rev(f4)`



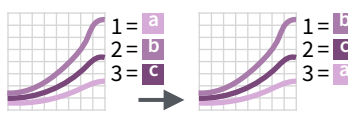
fct_shift(f) Shift levels to left or right, wrapping around end.
`fct_shift(f4)`



fct_shuffle(f, n = 1L) Randomly permute order of factor levels.
`fct_shuffle(f4)`

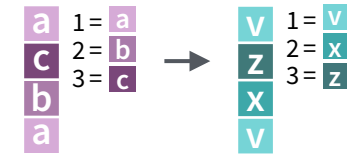


fct_reorder(f, x, fun = median, ..., desc = FALSE) Reorder levels by their relationship with another variable.
`boxplot(PlantGrowth, weight ~ fct_reorder(group, weight))`

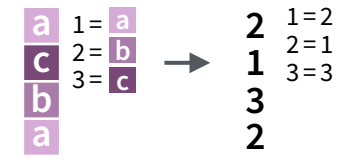


fct_reorder2(f, x, y, fun = last2, ..., desc = TRUE) Reorder levels by their final values when plotted with two other variables.
`ggplot(diamonds, aes(carat, price, color = fct_reorder2(color, carat, price))) + geom_smooth()`

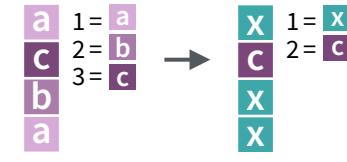
Change the value of levels



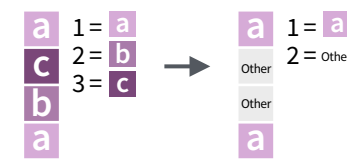
fct_recode(f, ...) Manually change levels. Also **fct_relabel()** which obeys purrr::map syntax to apply a function or expression to each level.
`fct_recode(f, v = "a", x = "b", z = "c")`
`fct_relabel(f, ~ paste0("x", .x))`



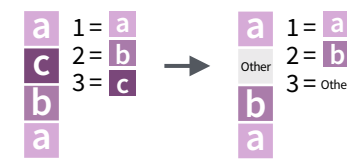
fct_anon(f, prefix = "")
Anonymize levels with random integers.
`fct_anon(f)`



fct_collapse(f, ..., other_level = NULL)
Collapse levels into manually defined groups.
`fct_collapse(f, x = c("a", "b"))`



fct_lump_min(f, min, w = NULL, other_level = "Other") Lumps together factors that appear fewer than min times. Also **fct_lump_n()**, **fct_lump_prop()**, and **fct_lump_lowfreq()**.
`fct_lump_min(f, min = 2)`

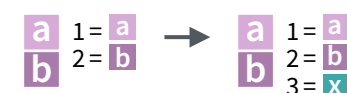


fct_other(f, keep, drop, other_level = "Other") Replace levels with "other."
`fct_other(f, keep = c("a", "b"))`

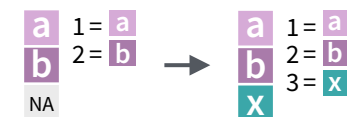
Add or drop levels



fct_drop(f, only) Drop unused levels.
f5 <- **factor**(c("a", "b"), c("a", "b", "x"))
f6 <- **fct_drop**(f5)



fct_expand(f, ...) Add levels to a factor.
`fct_expand(f6, "x")`



fct_na_value_to_level(f, level = "(Missing)") Assigns a level to NAs to ensure they appear in plots, etc.
f7 <- **factor**(c("a", "b", NA))
`fct_na_value_to_level(f7, level = "(Missing)")`